

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF
ENGINEERING)** Department of Artificial Intelligence and Machine Learning

ASSESSMENT 1 – Design and Develop Model (5 QUESTIONS)

Course Code:	MLA0407	Course Title:	Deep Learning Assessment 1: Design and Develop Model (5 Questions × 10 Marks)
CO Assessed:	CO3	Assessment:	
Weightage:	35% of CIA	Total Marks:	
CO1 Statement:	<i>Apply the fundamental concepts of machine learning and neural networks to design and analyze learning algorithms using perceptrons, multilayer perceptrons, forward propagation, backpropagation, gradient descent optimization techniques, and Maximum Likelihood Estimation for solving real-world problems.</i>		
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Section / Batch:	AIML	Date:	

Code of Conduct

I Anand Paul(192425207)(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: _____

QUESTIONS: 5 Total Marks: 50

Annexure A

Design and Develop Model

1. Transfer Learning for Image Classification

Design and develop an image classification model using Transfer Learning with DenseNet. Use a pre trained DenseNet model as a feature extractor, add a custom classification layer, and evaluate the model's performance on a suitable dataset.

2. Fine-Tuning a Pre-Trained Model

Develop an image classification system using a pre-trained deep learning model and apply fine tuning. Compare the performance of the model before and after fine-tuning and analyze the impact of unfreezing different layers.

3. Feature Extraction-Based Model

Design and develop a deep learning model using a pre-trained CNN as a feature extractor for a real world classification problem. Extract deep features, train a custom classifier, and evaluate the effectiveness of transfer learning.

4. Medical Image Classification

Design and develop a transfer learning-based model for classifying medical images, such as pneumonia detection, skin disease classification, or brain tumor detection. Use a pre-trained model and evaluate its accuracy and other performance metrics.

5. RNN for Text Classification

Design and develop a Recurrent Neural Network (RNN) model for text classification. Use an appropriate dataset to classify text into different categories and analyze how sequential information is processed by the network.

6. Sentiment Analysis Using LSTM

Develop a sentiment analysis system using Long Short-Term Memory (LSTM). Classify user reviews or social media posts as positive, negative, or neutral and compare the results with a traditional RNN model.

7. Bidirectional RNN for Sequence Classification

Design and develop a Bidirectional RNN (BiRNN) model for sequence classification. Compare its performance with a standard RNN and explain how processing information in both forward and backward directions improves the model.

8. Next-Word Prediction Using LSTM

Develop an LSTM-based next-word prediction system. Train the model on a suitable text corpus and demonstrate how the model learns sequential patterns and predicts the most probable next word.

9. Encoder–Decoder Sequence-to-Sequence Model

Design and develop an Encoder–Decoder-based Sequence-to-Sequence model for an application such as text summarization, machine translation, or question-answer generation. Explain the role of the encoder and decoder in processing sequences.

10. Comparative Deep Learning Model

Design and develop multiple deep learning models for the same problem using Transfer Learning, RNN, BiRNN, or LSTM techniques. Compare their architectures and performance using appropriate evaluation metrics, and justify which model is most suitable for the selected application.

Page 1 – Introduction

Title: Design and Development of Deep Learning Models

Introduction:

Deep Learning is a branch of Machine Learning that uses artificial neural networks to solve complex problems. It is widely used in image classification, text classification, sentiment analysis, medical diagnosis, and sequence prediction. This assessment focuses on different deep learning techniques such as Transfer Learning, DenseNet, RNN, LSTM, BiRNN, and Encoder–Decoder models.

Objectives:

- Understand different deep learning architectures.
 - Apply pre-trained models using Transfer Learning.
 - Perform image and text classification.
 - Understand sequential data processing using RNN and LSTM.
 - Compare different models using performance metrics.
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Page 2 – Transfer Learning Using DenseNet

Aim:

To develop an image classification model using a pre-trained DenseNet model.

Method:

1. Select an image dataset such as CIFAR-10.
2. Load the pre-trained DenseNet model.
3. Remove the original classification layer.
4. Freeze the pre-trained layers.
5. Add a new Dense and Softmax classification layer.
6. Train the model on the selected dataset.
7. Evaluate using accuracy and F1-score.

Architecture:

Input Image → DenseNet → Feature Extraction → Dense Layer → Softmax → Class

Result:

The pre-trained DenseNet model can classify images effectively while requiring less training time.

Page 3 – Fine-Tuning a Pre-Trained Model

Aim:

To improve image classification by fine-tuning a pre-trained deep learning model.

Method:

- Load ResNet50/DenseNet.
- Freeze all layers initially.
- Train the classification layer.
- Unfreeze the last few layers.
- Train again using a low learning rate.
- Compare the results.

Comparison:

Model	Training	Expected Performance
Frozen Model	Classifier only	Good
Fine-Tuned Model	Last few layers	Better

Result:

Fine-tuning allows the model to adapt pre-trained features to the new dataset and can improve classification performance.

Page 4 – Feature Extraction-Based Model

Aim:

To extract useful image features from a pre-trained CNN and use them for classification.

Method:

1. Load a pre-trained CNN such as VGG16.
2. Remove its final classification layer.
3. Input images into the CNN.
4. Extract deep feature vectors.
5. Train a classifier such as SVM or Dense layer.
6. Test the classifier.

Architecture:

Image → Pre-trained CNN → Deep Features → Classifier → Output

Advantages:

- Less training time.
- Requires less computational power.
- Uses powerful pre-learned features.

- Suitable for small datasets.

Result:

Deep feature extraction provides useful representations for image classification.

Page 5 – Medical Image Classification

Aim:

To develop a transfer-learning model for detecting pneumonia from chest X-ray images.

Dataset:

Chest X-ray dataset containing Normal and Pneumonia images.

Method:

- Resize X-ray images.
- Normalize pixel values.
- Load a pre-trained DenseNet/ResNet.
- Add a binary classification layer.
- Train and validate the model.
- Generate a confusion matrix.

Architecture:

X-Ray → CNN → Feature Extraction → Dense Layer → Normal/Pneumonia

Evaluation Metrics:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Result:

Transfer learning can provide an effective approach for medical image classification.

Page 6 – RNN for Text Classification

Aim:

To develop an RNN model for classifying text.

Dataset:

IMDB movie review dataset.

Method:

1. Collect text reviews.
2. Tokenize the words.
3. Convert words into numerical sequences.
4. Apply word embeddings.
5. Pass sequences through an RNN.
6. Use a Dense layer for classification.

Architecture:

Text → Tokenization → Embedding → RNN → Dense → Output

Result:

The RNN learns sequential relationships between words and classifies the text into appropriate categories.

Page 7 – Sentiment Analysis Using LSTM

Aim:

To classify reviews as positive or negative using LSTM.

Method:

- Use IMDB reviews.
- Tokenize the text.
- Convert words into sequences.
- Apply an embedding layer.
- Use an LSTM layer.
- Add a sigmoid output layer.
- Train and evaluate the model.

Architecture:

Review → Tokenization → Embedding → LSTM → Dense → Positive/Negative

RNN vs LSTM:

Feature	RNN	LSTM
Memory	Short-term	Long-term
Long sequences	Less effective	More

		effective
Vanishing gradient	More likely	Reduced
Performance	Good	Usually better

Result:

LSTM is effective for sentiment analysis because it can remember important information from earlier words.

Page 8 – Bidirectional RNN

Aim:

To develop a BiRNN model for sequence classification.

Concept:

A normal RNN processes information in one direction, while a BiRNN processes information in both directions.

Architecture:

Input Sequence → Forward RNN ↘

→ Output

Input Sequence → Backward RNN ↗

Advantages:

- Uses past information.
- Uses future information.
- Better understanding of context.
- Useful for text classification.

Comparison:

Model	Direction	Context
RNN	Forward	Previous words
BiRNN	Forward + Backward	Previous + future words

Result:

BiRNN can improve sequence classification by considering information from both directions.

A. Next-Word Prediction Using LSTM

Aim:

To predict the next word in a sentence.

Example:

Input:

Machine learning is

Predicted word:

powerful

Process:

Text Corpus → Tokenization → Sequences → LSTM → Softmax → Next Word

Result:

The LSTM learns patterns in the training text and predicts probable next words.

B. Encoder–Decoder Model

Application: Machine Translation

Architecture:

English Sentence → Encoder → Context → Decoder → French Sentence

Encoder: Converts the input sequence into a meaningful representation.

Decoder: Generates the output sequence one word at a time.

Result:

Encoder–Decoder models are useful for translation, summarization and question-answer generation.